A Spectrum Criterion for Eventually Positive Semigroups on $C(K)$

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Source Question

Daners–Glueck–Kennedy, *Eventually Positive Semigroups of Linear Operators*, arXiv:1511.09020, prove in Corollary 6.5 that a real uniformly eventually strongly positive C_0-semigroup on $C(K)$ has generator with non-empty spectrum. They then remark that for a positive semigroup this is true without irreducibility or strong positivity assumptions, and that it is unclear whether

$$\sigma(A) \neq \varnothing$$

holds for uniformly or individually eventually positive semigroups on $C(K)$ in general.

Partial Result

Theorem 1. *Let K be a non-empty compact Hausdorff space and let $(T(t))_{t\geq 0}$ be a real individually eventually positive C_0-semigroup on $C(K)$, with generator A. Suppose that for some $t_1 > 0$, $T(t_1)$ is a positive operator. Assume moreover that at least one of the following two minorization conditions holds:*

1. *There are $u \in C(K)$, $u \gg 0$, and $\varepsilon > 0$ such that*

$$T(t_1)u \geq \varepsilon u.$$

2. *There are a non-zero positive functional $\varphi \in C(K)'$ and $\varepsilon > 0$ such that*

$$T(t_1)'\varphi \geq \varepsilon\varphi.$$

Then $\sigma(A) \neq \varnothing$.

Proof Intuition

The source already supplies the hard spectral-bound theorem: on $C(K)$, for real individually eventually positive semigroups, the spectral bound equals the growth bound. Thus it is enough to rule out growth bound $-\infty$. A single positive tail operator with a lower bound on an order unit, or on a positive functional in the dual, does exactly that: its powers cannot decay faster than ε^n. This gives a finite lower bound for the semigroup growth, hence a finite spectral bound, hence non-empty spectrum.

Proof

We use Theorem 6.4 of Daners–Glueck–Kennedy: since the semigroup is real and individually eventually positive on $C(K)$,

$$s(A) = \omega_0(A),$$

where both quantities are understood in the extended sense.

Assume first that $T(t_1)u \geq \varepsilon u$ for some $u \gg 0$. Since $T(t_1)$ is positive, induction gives

$$T(nt_1)u = T(t_1)^n u \geq \varepsilon^n u \qquad (n \in \mathbb{N}).$$

Therefore

$$\|T(nt_1)\| \geq \frac{\|T(nt_1)u\|_\infty}{\|u\|_\infty} \geq \varepsilon^n.$$

It follows that the growth bound satisfies

$$\omega_0(A) \geq \frac{\log \varepsilon}{t_1} > -\infty.$$

By the source theorem, $s(A) = \omega_0(A) > -\infty$. If $\sigma(A)$ were empty, then by definition $s(A) = \sup\{\operatorname{Re}\lambda : \lambda \in \sigma(A)\} = -\infty$, a contradiction.

The dual condition is identical. Since $T(t_1)$ is positive, $T(t_1)'$ is positive, and

$$T(nt_1)'\varphi = T(t_1)'^n\varphi \geq \varepsilon^n\varphi.$$

For a positive functional on $C(K)$, $\|\varphi\| = \varphi(\mathbf{1}) > 0$; hence

$$\|T(nt_1)\| = \|T(nt_1)'\| \geq \frac{\|T(nt_1)'\varphi\|}{\|\varphi\|} \geq \varepsilon^n.$$

The same growth-bound argument proves $\sigma(A) \neq \varnothing$.

Corollary 1. *Let $(T(t))_{t\geq 0}$ be a real uniformly eventually positive C_0-semigroup on $C(K)$. If there are t_1 beyond the eventual positivity threshold, $u \gg 0$, and $\varepsilon > 0$ such that*

$$T(t_1)u \geq \varepsilon u,$$

then the generator has non-empty spectrum. The same conclusion holds under the dual minorization $T(t_1)'\varphi \geq \varepsilon\varphi$ for a non-zero positive functional φ.

Scope and Novelty Check

This is a partial result only. It does not settle the full source remark, because a uniformly eventually positive semigroup might fail to minorize an order unit at every positive tail time. The cheap run indexes were searched for arXiv:1511.09020 and for the core phrases "eventually positive", "positive semigroups", "spectral bound", and "spectrum"; no existing packet for this exact spectrum criterion was found. The nearby packet for arXiv:1511.05294 concerns a different resolvent-versus-semigroup implication.

References

References

[1] D. Daners, J. Glueck, and J. B. Kennedy, *Eventually Positive Semigroups of Linear Operators*, J. Math. Anal. Appl. 433 (2016), 1561–1593; arXiv:1511.09020.